Distractors are the Real Enemy: Visual Distraction Adversarial Perturbations on SAM2-Based Video Tracking and Segmentation

Adonay Demewez Gebremedhin[1] Mohamad Alansari[2] Wessam Shehieb[1] Muzammal Naseer[2,3]

[1]Ajman University, Ajman, UAE [2]Khalifa University, Abu Dhabi, UAE
[3]University of Western Australia, Australia

Abstract

Promptable video trackers such as SAM2 achieve accurate and robust performance through strong temporal consistency, making them important foundations for downstream video understanding and tracking applications. As their use continues to expand, understanding their robustness to adversarial perturbations becomes increasingly important. Existing attacks on SAM2 largely focus on local mask degradation and short-term temporal feature disruption, leaving its longer-term dynamics and competing object interferences (Distractors) insufficiently explored. We observe that SAM2 itself exposes useful distractor cues when tracking becomes uncertain. Based on these observations, we propose ***DistractAttack****, a white-box attack that uses background distractors to disrupt target tracking persistently. Its core procedure,* ***Counterfactual Distractor Search*** *(CDS), propagates clean and attacked trajectories in parallel to discover plausible competing regions and identify those that become increasingly effective under perturbation. The discovered evidence is accumulated over time to weaken target support and promote recurrent alternatives or disappearance predictions. Extensive experiments under video-level and one-shot perturbations across Visual Object Tracking and Video Object Segmentation demonstrate strong attack effectiveness, raising DRE to* $4.4\times$ *and NRE to* $7.3\times$ *their nearest baselines under video perturbation, while transferring effectively across model scales and tracker variants in black-box settings. Project page:* https://adonaydem.github.io/sam2-da-page/.

1. Introduction

Recent advances in promptable video segmentation have demonstrated strong and generalizable performance across downstream tasks such as Visual Object Tracking (VOT) and Video Object Segmentation (VOS). SAM2 [33], in particular, extends promptable segmentation to video through a recurrent memory mechanism that propagates object information across frames. Building on this foundational model, recent methods have further improved long-term tracking through stronger motion modeling, memory management, and distractor handling [4, 9, 37, 41, 42]. While promptable trackers have become increasingly robust, it remains unclear whether their specialized designs resist adversarial perturbations or share transferable vulnerabilities.

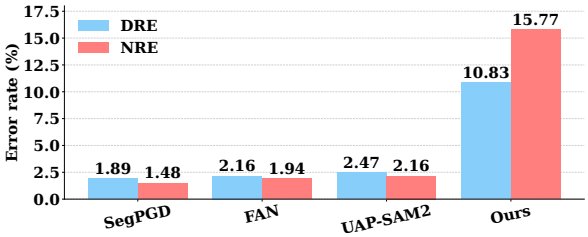


Figure 1. Drift-Rate Error (DRE) vs Not-Reported Error (NRE) performance across video perturbation methods. Higher DRE and NRE represent a balanced and comprehensive attack success.

Attacks against single-object tracking and VOS manipulate classification, bounding-box regression, temporal consistency, trajectories, and motion cues [5, 6, 13, 14, 18, 25], or exploit initialization errors and hard regions to induce persistent segmentation failure [19, 24]. Although these studies show that temporal recurrence and stored representations can amplify brief perturbations, their objectives were developed largely for earlier architectures such as Siamese networks [25] and do not readily transfer to foundation models such as SAM2. In parallel, attacks on the Segment Anything Model (SAM) [22] consider mask manipulation, universal and prompt-agnostic perturbations, and feature-specific objectives [1, 15, 17, 20, 26, 28, 32, 35, 38, 39, 43, 45, 46], but remain image-based and overlook the memory-driven dynamics of promptable video trackers. For SAM2, Robust-SAM showed that adversarial perturbations can disrupt temporal consistency [27],

while UAP-SAM2 jointly perturbs semantic features and consecutive-frame consistency [47]. BadVSFM further considers training-time backdoors [44]. Yet existing SAM2 attacks still largely target segmentation degradation or mask disappearance, while their temporal objectives remain centered on relatively local feature interactions rather than the long-term object trajectory. This leaves open how an attack can exploit the longer-term mechanisms that modern promptable trackers use to preserve and recover target identity under adverse challenges.

We introduce **DistractAttack**, a distractor-centric white-box attack for recurrent promptable segmentation that uses model-supported distractors to destabilize target tracking rather than merely erase the current mask. Its core, **Counterfactual Distractor Search** (CDS), propagates clean and attacked trajectories in parallel: the clean trajectory identifies plausible non-target competitors, while the attacked trajectory reveals which become increasingly influential under a bounded ℓ_∞ perturbation. Accumulating this evidence over time, CDS turns local ambiguity into persistent tracking failures, including wrong-object capture, spatial displacement, unstable localization, and disappearance. We evaluate video-level and one-shot perturbations across VOT and VOS, with transfer studies across model scales and state-of-the-art tracker variants. SAM2's memory mechanism is particularly vulnerable to one-shot initialization perturbations, whose effects persist throughout the video. Under video-perturbation and matched protocol, DistractAttack achieves the highest Drift-Rate Error (DRE) and Not-Reported Error (NRE) [23] by a substantial margin (see Fig. 1).

Our core contributions are summarized as follows:

- We propose the first distractor-centric adversarial attack against SAM2, exposing its vulnerability to distractors in the surrounding scene. **Counterfactual Distractor Search** couples clean and attacked trajectories to exploit evolving distractor evidence and induce persistent tracking failure.
- We extensively study DistractAttack under two complementary perturbation settings: **Video Perturbation (VP)** and **One-shot Perturbation (OP)** across both VOT and VOS tasks.
- We provide a broad evaluation against existing attacks and assess real-world practicality of our method via black-box transferability across model sizes and SAM2 tracker variants.

2. Related Work

2.1. Promptable Video Segmentation and Tracking

SAM introduced promptable class-agnostic image segmentation [22], and SAM2 extended this paradigm to video through a streaming memory that propagates an initialized object across frames [33]. SAM2 further broadened the family toward concept-prompted detection, segmentation, and tracking [3]. Recent SAM2 extensions strengthen long-term tracking through motion-aware memory selection in SAMURAI [42], multiple temporal hypotheses in SAM2Long [9], distractor-aware memory in DAM4SAM [37], position prompting in SAMITE [41], and combined long- and short-term modeling in HiM2SAM [4]. Their shared focus on distractors and memory reliability identifies *identity confusion* as a central challenge in promptable tracking. We study the complementary adversarial question of whether a bounded perturbation can exploit such distractors to progressively confuse the tracker's target representation and disrupt its temporal consistency.

2.2. Adversarial Video Tracking and Segmentation

Adversarial video tracking has shown that recurrence, association, and stored representations can turn brief perturbations into persistent errors. In visual object tracking, FAN, IoU Attack, SPARK, CSA, and ABA attack temporal box consistency, sparse online updates, classification responses, and motion blur [5, 13, 14, 18, 25], while adversarial VOS exploits initialization errors that aim to induce mask degradation [19, 24]. However, these frameworks remain closely tied to their victim architectures and have limited compatibility with recent SAM-based trackers. Attacks on SAM have covered mask removal, targeted generation, prompt-agnostic feature manipulation, region suppression and spatial–frequency attacks [15, 17, 20, 26, 28, 32, 35, 38, 39, 43, 45, 46]. Direct studies of SAM2 include Robust SAM (CPA), UAP-SAM2, BadVSFM, and DarkLLM [27, 36, 44, 47]. Among these methods, UAP-SAM2 alone introduces an explicit temporal objective, yet it merely relates consecutive encoder features rather than the history accumulated in recurrent memory. We aim to exploit accumulated target history and recurring distractor evidence to induce persistent confusion between the target and plausible competing regions.

3. Preliminary

Let $x_{1:T}$ denote a video, M_0 the target annotation provided only at the first frame, and h_t the recurrent tracker state at t. We follow standard single-object VOT and VOS protocols, where the target is initialized by a box and mask, respectively. SAM2 then performs:

$$(M_t^s, h_t, \mathcal{A}_t) = \Phi(x_t, h_{t-1}, M_0), \tag{1}$$

where M_t^s is the selected mask and $\mathcal{A}_t$ collects auxiliary decoder outputs, including confidence and predicted-IoU scores. Here, Φ may be either a clean or attacked tracker, Φ^c or Φ^a, with encoder features F_t^c or F_t^a, respectively. The SAM2 decoder also produces multiple candidate masks be-

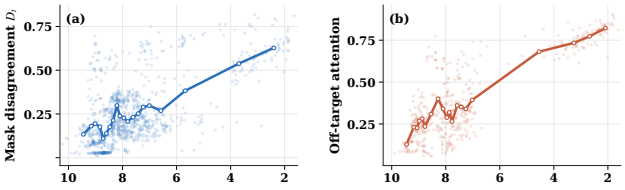


Figure 2. **Decoder ambiguity provides useful attack cues.** Lower objectness confidence (x-axis) is associated with (a) greater mask disagreement D_t and (b) stronger off-target attention mass. The lines show averages across bins.

fore selecting M_t^s. We refer to these candidates as **Mask Hypotheses** throughout the paper.

We assume a white-box attack with access to the model code, parameters, auxillary ouputs and gradients, and can collect publicly available datasets. The attack acts exclusively through bounded perturbations to the RGB input,

$$\widetilde{x}_t = \mathrm{clip}(x_t + \delta_t), \qquad \|\delta_t\|_\infty \le \varepsilon, \tag{2}$$

where ε denotes the perturbation budget.

4. Methodology

4.1. Observations and Overview

Observation I: Mask hypotheses diverge as confidence falls. At each frame, the SAM2 decoder produces several mask hypotheses and selects one using their predicted-IoU scores. We measure their disagreement as

$$D_t = 1 - \frac{1}{N_c} \sum_{(i,j)\in\mathcal{C}_t} \mathrm{IoU}\left(M_t^{(i)}, M_t^{(j)}\right), \qquad N_c = |\mathcal{C}_t|, \tag{3}$$

where $\mathcal{C}_t$ contains all pairs of predicted masks. A larger D_t therefore indicates greater disagreement. For this analysis, we run clean SAM2.1-L on ten randomly selected videos, collect frame-wise decoder outputs. As shown in Fig. 2(a), disagreement increases as confidence in the selected output falls. Further analysis shows that SAM 2 often shifts to distractors prior to tracking failure. The discarded hypotheses therefore provide model-generated proposals for regions that may compete with the target.

Observation II: Decoder attention moves away from the target as confidence falls. The selected mask may remain stable even as attention shifts toward competing regions. We measure the fraction of mask-token attention outside the target object. Fig. 2(b) shows that this fraction rises as confidence falls, indicating growing support outside the target. The attention map therefore complements the alternative masks in Obs. I by identifying non-target regions that already influence the decoder.

Together, these observations motivate *Counterfactual Distractor Search* (CDS), which separates evidence collection from perturbation optimization. A clean trajectory provides target features and competing regions, while an attacked trajectory reveals how distractor evidence evolves under the bounded perturbation. Optimization then suppresses target support, biases foreground responses toward discovered distractors, and corrupts target distinctiveness across multiple granularities. We consider two settings. **Video Perturbation (VP)** optimizes one perturbation per video, while **One-shot Perturbation (OP)** perturbs only the prompted frame [6, 19]. DistractAttack is primarily classified as untargeted because no distractor identity is specified in advance. Its distractor proposals guide the optimization but may instead exhibit wrong-object capture, spatial displacement, unstable switching, or disappearance.

4.2. Counterfactual Distractor Search (CDS)

4.2.1. Target Prototype Bank

For an arbitrary feature map F and region mask R, we define a *Prototype Generator*:

$$\psi(F, R) = \frac{1}{|R|} \sum_{u\in R} F(u). \tag{4}$$

where u indexes spatial locations within region R.

We propagate a clean trajectory and store a reliable clean history of the target in a lightweight prototype bank $\mathcal{T}$. Each reliable clean encoder features F_t^c with selected mask M_t^s contribute a foreground–background pair using the prototype generator,

$$p_t^+ = \psi(F_t^c, M_t^s), \qquad p_t^- = \psi(F_t^c, 1 - M_t^s). \tag{5}$$

The first frame pair is always retained. Later pairs are admitted only when the model's predicted-IoU score and objectness logit exceed 0.5 and 0, respectively. To avoid repeatedly storing near-duplicate target appearances, the resulting foreground prototype is further required to pass a cosine novelty gate:

$$\begin{aligned} &\max_{m\in\mathcal{T}} \cos\left(p_t^+, p_m^+\right) < \tau_{\mathrm{nov}} \\ \Longrightarrow \quad &\mathcal{T} \leftarrow \mathcal{T} \cup \{(p_t^+, p_t^-)\}. \end{aligned} \tag{6}$$

We set τ_{nov} to 0.98. These mechanisms help us reject unreliable and redundant updates while preserving appearance changes needed to distinguish the target from its surroundings.

4.2.2. Search Components

During video propagation, CDS searches for plausible distractors from a set of signals exposed by the clean and attacked trajectories. We consider three sources of evidence.

A. Semantic Distractor Search. We use the target prototype bank $\mathcal{T}$ and the current clean encoder feature map F_t^c to measure whether each spatial location resembles the target more than its surrounding background.

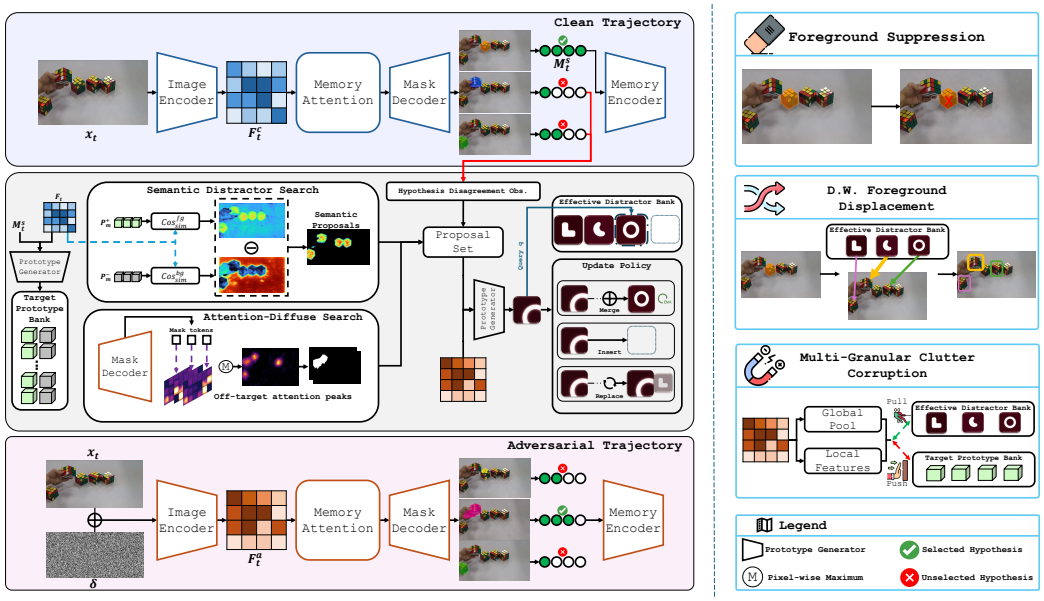


Figure 3. **Overview of DistractAttack.** (a) Counterfactual Distractor Search: Clean and attacked trajectories provide counterfactual target references for proposing, validating, and accumulating distractors. (b) CDS evidence guides foreground suppression, multi-granular clutter corruption, and distractor-weighted foreground displacement to induce persistent tracking failure.

For each stored target prototype m, $d_m(u)$ measures how much location u resembles the target more than the background:

$$d_m(u) = \Big[\cos\big(F_t^c(u), p_m^+\big) - \cos\big(F_t^c(u), p_m^-\big) \Big]_+ . \quad (7)$$

We then normalize each d_m by its maximum response and average across stored target prototypes:

$$S_t(u) = \frac{1}{|\mathcal{T}|} \sum_{m \in \mathcal{T}} \frac{d_m(u)}{\max_v d_m(v) + \epsilon}. \quad (8)$$

S_t provides a spatial map of regions that remain similar to the target across its stored appearances [41]. We mask the selected target using M_t^s and retain the remaining spatially distinct high-response regions as semantic distractor proposals, $\mathcal{P}_t^{\text{sem}}$.

Proposals in $\mathcal{P}_t^{\text{sem}}$ capture plausible appearance-based distractors, but not the decoder's evolving memory or longer-term temporal behavior. Following Observations I and II, we therefore examine SAM2's tracking behavior to identify emerging mask-decoder vulnerabilities.

B. Hypothesis Disagreement Observation. CDS compares the clean decoder's alternative mask hypotheses with the selected mask M_t^s. If an alternative predicts a distinct region outside M_t^s, we retain that region as an alternative distractor proposal in the set $\mathcal{P}_t^{\text{alt}}$. These proposals are particularly informative because they expose alternative regions the decoder already considers plausible. However, they may be unavailable when SAM2 is confident, leading to its mask hypotheses largely agreeing.

C. Adversarial Attention-Diffuse Search. CDS further examines mask-token attention along a parallel attacked trajectory. We find that this attention begins to spread toward off-target regions before the predicted mask visibly fails. After scaling each attention map by its uniform spatial expectation, we define

$$H_t(u) = \max_k \log\big(A_{t,k}^a(u)\big), \quad (9)$$

where $A_{t,k}^a$ denotes the attention of mask token k outside the target. Positive values of H_t therefore indicate stronger attention. The resulting off-target regions enter the attention diffuse set $\mathcal{P}_t^{\text{att}}$.

4.2.3. Effective Distractor Bank

The three searches provide complementary distractor evidence in the form of binary masks. After removing overlapping regions, CDS combines all available proposals into

$$\mathcal{P}_t = \mathcal{P}_t^{\text{sem}} \cup \mathcal{P}_t^{\text{alt}} \cup \mathcal{P}_t^{\text{att}}. \quad (10)$$

Much like the target prototype bank $\mathcal{T}$, an *Effective Distractor Bank* $\mathcal{E} = \{e_s\}_{s=1}^K$ admits a compact history of competing appearances using $\mathcal{P}_t$ and the adversarial encoder feature F_t^a. Unlike the target prototype bank, whose identity is initiated by the initial prompt, these proposals carry no annotation anchors. Naive updates may therefore split one distractor across multiple slots or merge distinct distractors.

Algorithm 1. Effective Distractor Bank Update Policy

Require: $\mathcal{P}_t$: distractor proposal set, F_t^a: attacked feature map, $\mathcal{E} = \{e_s\}$: effective distractor bank
Require: K: bank capacity, θ_{merge}: merge threshold, η: EMA update rate
for $P_t^j \in \mathcal{P}_t$ **do**
  Form a prototype: $q \leftarrow \psi(F_t^a, P_t^j)$
  Find its closest pair in $\mathcal{E}$: $i \leftarrow \arg\max_s \cos(q, e_s)$
  if $\cos(q, e_i) \geq \theta_{\text{merge}}$ **then**
    Merge: $e_i \leftarrow (1-\eta)e_i + \eta q$
  else if $|\mathcal{E}| < K$ **then**
    Insert q into $\mathcal{E}$
  else
    Replace with the least used slot.
  end if
end for

We use conservative association and updates to maintain stable histories.

Effective Distractor Bank Update Policy. Each proposal in $\mathcal{P}_t$ forms a prototype from gradient-detached F_t^a for comparison with occupied slots in $\mathcal{E}$. If its closest slot is sufficiently similar, the two are merged through an exponential moving average (EMA) [34]. Otherwise, the candidate fills an empty slot or replaces the least-used one when the bank is full. A least used slot is the one that has the least EMA update. This turns repeated evidence into a compact set of stable distractors, preventing the bank from growing with redundant regions and focusing the attack on a few *effective* alternatives (See Algorithm 1).

4.3. Attack Objectives

The attack objective set is designed to induce persistent tracking failure. Using the target prototype and effective distractor banks accumulated over time, it suppresses foreground support on the target, promotes CDS-selected competing regions, and corrupts target distinctiveness along the attacked trajectory.

Foreground suppression. Given the attacked mask M_t^a and the clean predicted target M_t^s from the parallel clean trajectory, we first suppress the attacked prediction M_t^a over the clean target M_t^s:

$$\mathcal{L}_{\text{mask}} = \text{BCE}_{M_t^s}\left(M_t^a, \mathbf{0}\right), \tag{11}$$

where BCE_R evaluates binary cross entropy (BCE) only within R. Unlike objectives that also suppress the background [43, 47], this leaves off-target regions available to receive foreground support.

Distractor-weighted foreground displacement. Foreground suppression removes support from the target but does not specify where it should move. A complete displacement objective can instead promote foreground to be over the active CDS proposals:

$$\mathcal{L}_{\text{disp}} = \frac{1}{|\mathcal{P}_t|} \sum_{j \in \mathcal{P}_t} \text{BCE}_{P_t^j}\left(M_t^a, \mathbf{1}\right), \tag{12}$$

where $\mathcal{P}_t$ is the current distractor proposal set. Treating all proposals equally, however, ignores which regions repeatedly expose the strongest vulnerabilities along the attacked trajectory [29]. Promoting multiple unrelated regions also conflicts with the tracker's single-target nature and weakens the attack. We therefore leverage the effective distractor bank which preserves evidence from the strongest proposal trajectories, providing long-term precedent and a coherent attack direction. For each proposal P_t^j, we pool $z_t^j = \psi(F_t^a, P_t^j)$ and define a_t^j as its maximum cosine similarity to the occupied distractor slots in $\mathcal{E}$. We then apply softmax:

$$w_t^j = \frac{e^{\,a_t^j/\tau_w}}{\sum_{q \in \mathcal{P}_t} e^{\,a_t^q/\tau_w}}, \qquad j \in \mathcal{P}_t, \tag{13}$$

so that w_t^j concentrates foreground on proposals most strongly supported by the accumulated distractor evidence. τ_w denotes temperature. The resulting enhanced objective is

$$\mathcal{L}_{\text{disp}} = \sum_{j \in \mathcal{P}_t} w_t^j \ \text{BCE}_{P_t^j}\left(M_t^a, \mathbf{1}\right). \tag{14}$$

Multi-granular clutter corruption. Mask displacement biases foreground support toward competing regions, but the target may remain sufficiently distinct in feature space, deeming it sufficient enough for SAM2 to continue tracking the true target. We therefore contrast attacked feature maps against the effective distractor bank $\mathcal{E}$ and the clean target foreground prototypes $\mathcal{T}^+ = \{p_m^+\}$.

For an arbitrary feature q, we directly define a multi-positive contrastive loss as

$$\mathcal{L}_{\text{con},q} = -\log \frac{\sum_{e \in \mathcal{E}} e^{\cos(q,e)/\tau_c}}{\sum_{e \in \mathcal{E}} e^{\cos(q,e)/\tau_c} + \sum_{p \in \mathcal{T}^+} e^{\cos(q,p)/\tau_c}}. \tag{15}$$

Minimizing this loss pulls the representation toward accumulated distractors while pushing it away from the clean target history [7, 21]. τ_c denotes contrastive temperature.

From F_t^a, we form a global query q_t^G by average pooling and local queries q_t^u from its spatial features, and evaluate the contrast at both granularities. The resulting objective is

$$\mathcal{L}_{\text{clut}} = \eta_G \, \mathcal{L}_{\text{con},q_t^G} + \eta_L \sum_u \alpha_t^u \, \mathcal{L}_{\text{con},q_t^u}. \tag{16}$$

where α_t^u is weight over target-region features, focusing the local contrastive loss on the target region to more effectively weaken its representation. The complete objective is

$$\mathcal{L} = \lambda_M \mathcal{L}_{\text{mask}} + \lambda_D \mathcal{L}_{\text{disp}} + \lambda_C \mathcal{L}_{\text{clut}}. \tag{17}$$

where λ_M, λ_D, λ_C are loss coefficients. Using the computed gradient $g_{\mathcal{W}}$, the perturbation is a signed gradient update [11, 30]:

$$\delta \leftarrow \Pi_{\|\delta\|_\infty \le \epsilon} \left(\delta - \alpha \operatorname{sign}\left(g_{\mathcal{W}}\right)\right), \tag{18}$$

where α is the step size and the scope of δ follows one of the following perturbation settings:

- **Video Perturbation (VP):** A separate perturbation δ_v is optimized for each video v. Once optimized, the same perturbation is applied across all frames of that video.
- **One-shot Perturbation (OP):** We consider a more practical setting in which the adversary can manipulate only the initially prompted frame [6, 19]. The attacked sequence is defined as

$$\widetilde{x}_t = \begin{cases} \text{clip}(x_0 + \delta_o), & t = 0, \\ x_t, & t > 0, \end{cases} \qquad \|\delta_o\|_\infty \le \epsilon. \tag{19}$$

  At each optimization step, clean and attacked trajectories are initialized from the same prompt and propagated over a rollout of N frames to feed banks $\mathcal{T}$ and $\mathcal{E}$. Here, N defines only the observation horizon for bank construction and introduces no additional prompts, annotations, or perturbations to later frames. Both banks are detached and used as temporal context for the attack objective, while gradients are propagated only to δ_o. Once optimized, δ_o is applied once to the prompted frame, after which the tracker proceeds on the remaining clean frames without further adversarial intervention.

Importantly, in both settings, only the initial mask and video frames are known to the adversary.

5. Experiments

Setup protocol. We evaluate VP and OP on visual object tracking (VOT) and video object segmentation (VOS). For VOT, we use LaSOT [10], UTB180 [2], GOT-10k [16], and LaSOT-ext [10], including the distractor-distilled DiDi [37]. Each is initialized by the ground-truth box. Following VOTS [23], we macro-average tracking quality Q, accuracy Acc, robustness Rob, DRE, and NRE across sequences. For VOS, we use DAVIS 2017 (DAVIS) [31], YouTube-VOS 2018 (YT) [40], MOSEv2 [8], and the SA-V test set [33]. The ground-truth mask initializes frame 0, and we report the standard $\mathcal{J}\&\mathcal{F}$. During optimization, OP may use $N \in \{1, 10, 20\}$ rollout frames, while its evaluation is on the complete sequence. We use the official SAM2.1 checkpoint and none of the dataset sequences were used to train it. VP uses $\epsilon = 8/255$ and $\alpha = 2/255$ while OP uses $\epsilon = 4/255$ and $\alpha = 1/255$. Both use 10 iterations, and a weight of 1.0 for all loss terms. We cap $\mathcal{T}$ and $\mathcal{E}$ at 20 and 10 prototypes, respectively, and set θ_{merge} 0.85 and the EMA rate η 0.1. Single-image and universal

Table 1. Video (VP) and One-shot (OP) attack perturbation mechanisms on SAM2.1-L. VOS results are given in $\mathcal{J}\&\mathcal{F}$. Arrows indicate a stronger attack.

Setting	Method	VOT					VOS				
		$Q\uparrow$	Acc.↓	Rob.↓	DRE%↑	NRE%↑	DAVIS↓	YT↓	MOSEv2↓	SA-V↓	Avg.↓
VP	Clean inference	0.836	0.842	0.989	0.68	0.42	91.92	86.66	82.57	87.43	87.15
	AttackSAM [43]	0.798	0.807	0.981	1.10	0.83	80.34	75.92	72.88	77.10	76.56
	DarkSAM	0.772	0.784	0.976	1.38	1.05	77.42	70.18	69.55	73.84	72.75
	CPA [27]	0.756	0.769	0.972	1.56	1.22	74.90	67.62	66.81	71.33	70.17
	SegPGD [12]	0.735	0.748	0.966	1.89	1.48	72.88	64.95	65.72	69.04	68.15
	UAD [28]	0.718	0.732	0.962	2.04	1.70	71.56	63.72	64.40	67.85	66.88
	FAN [25]	0.701	0.720	0.959	2.16	1.94	69.31	60.44	61.87	65.26	64.22
	UAP-SAM2 [47]	0.689	0.709	0.954	2.47	2.16	62.81	51.37	62.55	57.94	58.67
	DistractAttack	**0.420**	**0.554**	**0.734**	**10.83**	**15.77**	**59.38**	**44.46**	**49.42**	**50.62**	**50.97**
OP	DarkSAM	0.803	0.811	0.971	1.15	1.78	61.48	53.21	58.75	56.84	57.57
	AttackSAM [43]	0.816	0.822	0.976	1.04	1.36	66.12	57.44	62.08	60.35	61.50
	CPA [27]	0.791	0.802	0.865	1.29	12.21	56.72	47.63	54.30	51.18	52.46
	SegPGD [12]	0.775	0.787	0.952	1.52	3.26	50.83	41.57	48.66	45.21	46.57
	UAD [28]	0.764	0.778	0.668	1.88	31.31	47.35	38.72	45.19	42.60	43.47
	FAN [25]	0.354	0.369	0.520	2.10	45.87	43.92	34.88	42.75	39.46	40.25
	UAP-SAM2 [47]	0.245	0.266	0.481	1.39	50.54	36.84	26.95	37.88	31.75	33.36
	ARA-Attack [24]	0.213	0.235	0.423	2.74	54.92	32.14	22.81	34.57	28.93	29.61
	VOSPGD [19]	0.172	0.193	0.281	2.55	**69.45**	19.43	16.59	37.31	24.73	24.52
	DistractAttack$_{N=1}$	0.131	0.148	0.270	8.71	64.34	16.45	11.12	33.34	27.09	22.00
	DistractAttack$_{N=10}$	0.101	0.126	0.278	9.09	63.14	16.11	10.12	31.24	25.76	20.81
	DistractAttack$_{N=20}$	**0.076**	**0.115**	**0.250**	**9.28**	65.73	**15.85**	**9.38**	**31.02**	**21.99**	**19.56**

baselines retain their defining objectives and mechanisms when adapted to sample-wise VP and OP. Stronger attacks correspond to lower Q, accuracy, robustness, and $\mathcal{J}\&\mathcal{F}$, and higher DRE and NRE. Experiments run on an NVIDIA A100 80GB GPU. Further protocol and baseline adaptation details are provided in Appendix 8.

Comparison against other methods. We compare image-segmentation, tracking, and VOS attacks on SAM2.1-Large in Tab. 1. ARA-Attack and VOSPGD are reported only under OP because they perturb the annotated initialization frame [19, 24]. DistractAttack is strongest overall in both settings. Under VP, it raises DRE from UAP-SAM2's 2.47% to 10.83%. Under OP, N denotes the rollout used to construct the target and distractor banks. At $N = 1$, CDS uses only initialization-frame evidence and lowers Q from 0.172 to 0.131 against VOSPGD. Increasing N to 20 adds

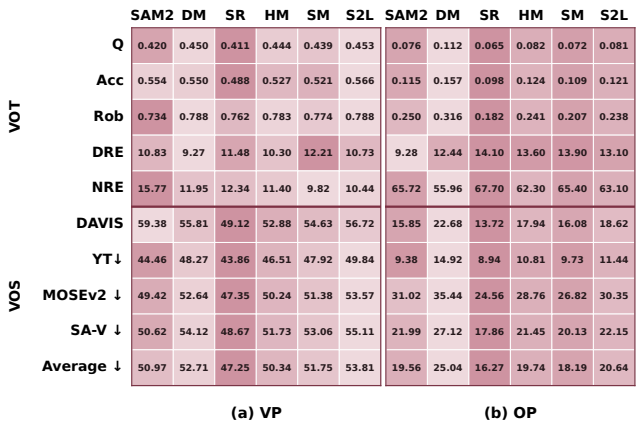
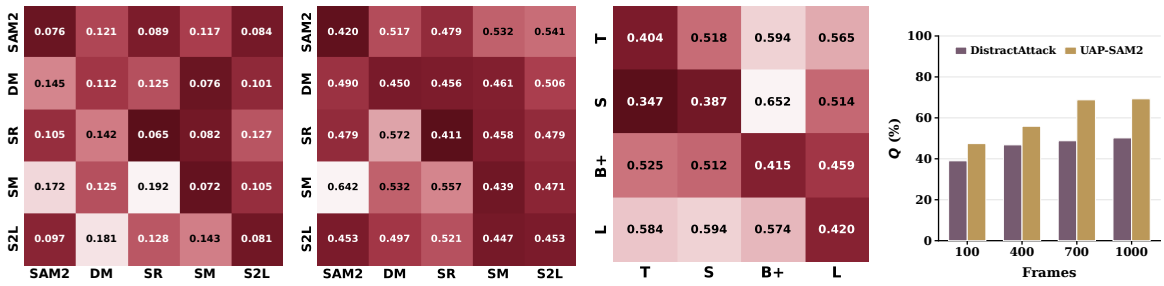


Figure 4. **DistractAttack across tracker variants** under (a) VP and (b) OP. DM, SR, HM, SM, and S2L denote DAM4SAM, SAMURAI, HiM2SAM, SAMITE, and SAM2Long. VOS entries report $\mathcal{J}\&\mathcal{F}$.



(a) Cross-variant OP (b) Cross-variant VP (c) Cross-size (d) Cross-time

Figure 5. **Black-box transferability of DistractAttack.** Panels show cross-variant transfer under (a) OP and (b) VP, (c) cross-size transfer among SAM2.1 models, and (d) zero-shot temporal transfer. Rows denote source models and columns denote targets. Lower off-diagonal Q indicates stronger transfer. In (d), perturbations optimized on frames 0–100 are evaluated on longer windows

unannotated frames and lowers the VOS average by a further 2.44 points, showing the benefit of temporal context. OP propagates severe initialization errors through memory, yielding more empty masks and higher NRE, while VP sustains misdirection and higher DRE. These complementary failure modes are also visible in Fig. 6.

DistractAttack across tracker variants. We evaluate DistractAttack on SAMURAI [42], DAM4SAM [37], HiM2SAM [4], SAMITE [41], and SAM2Long [9], all based on SAM2.1-L. Under VP, SAM2Long and DAM4SAM are slightly more resistant, while SAMURAI is most affected. OP with $N = 20$ is stronger across every variant. DAM4SAM remains the most resistant but still degrades sharply. Specialized distractor and memory mechanisms therefore offer only partial protection against tracking and segmentation failures. Across SAM2.1 sizes, Q falls to 0.387–0.420 with only modest variation across scales (Fig. 5).

Black-box transferability. Since DistractAttack uses white-box optimization, we assess its applicability in the black-box setting. Following standard transfer-based black-box evaluation [19, 25, 28], each perturbation is optimized on a source model and applied unchanged to a target. Fig. 5 evaluates four complementary axes. Cross-variant transfer among SAM2.1, DAM4SAM, SAMURAI, SAMITE, and SAM2Long yields off-diagonal Q ranges of 0.076–0.192 under OP and 0.447–0.642 under VP. All variants use SAM2.1-L, with $N = 20$ for OP. Transfer across the T, S, B+, and L sizes also remains effective, showing generalization across scales. For long-horizon transfer, perturbations optimized over frames 0–100 on ten randomly selected samples are evaluated unchanged at frames 400, 700, and 1000. On SAM2.1-L, Q increases by 28.7% for DistractAttack versus 46.1% for UAP-SAM2, supporting the long-term effect of our trajectory-focused optimization over the consecutive-frame optimization of UAP-SAM2. Complementary VOS transfer is provided in Appendix 9.2.

Table 2. Ablation of DistractAttack on SAM2.1-L under video (VP) and one-shot (OP) perturbations.

(a) Attack objectives and optimization.

Setting	$\mathcal{L}_{\text{mask}}$	$\mathcal{L}_{\text{clut}}$	$\mathcal{L}_{\text{disp}}$	VP $Q\downarrow$	VP DRE $\uparrow$	VP VOS Avg.$\downarrow$	OP $Q\downarrow$	OP DRE $\uparrow$	OP VOS Avg.$\downarrow$
Attack objectives									
$\mathcal{L}_{\text{mask}}$ only	✓			0.631	2.78	68.42	0.497	0.78	47.83
w/o $\mathcal{L}_{\text{mask}}$		✓	✓	0.561	9.89	62.76	0.244	11.96	34.12
w/o $\mathcal{L}_{\text{disp}}$	✓	✓		0.502	6.99	58.61	0.291	3.89	37.64
w/o $\mathcal{L}_{\text{clut}}$	✓		✓	0.491	7.78	56.94	0.202	4.05	30.87
Full	✓	✓	✓	0.420	10.83	50.97	0.076	9.28	19.56
Iteration length									
5 passes	✓	✓	✓	0.501	5.77	58.87	0.119	8.78	24.83
10 passes	✓	✓	✓	0.420	10.83	50.97	0.076	9.28	19.56
15 passes	✓	✓	✓	0.343	13.67	49.84	0.071	7.45	20.41
Perturbation magnitude									
$\epsilon = 4/255$	✓	✓	✓	0.562	10.44	62.03	0.076	9.28	19.56
$\epsilon = 8/255$	✓	✓	✓	0.420	10.83	50.97	0.069	8.49	18.74
$\epsilon = 16/255$	✓	✓	✓	0.390	9.51	49.91	0.046	9.95	17.82

(b) CDS evidence and temporal design.

Setting	$\mathcal{P}_t^{\text{sem}}$	$\mathcal{P}_t^{\text{alt}}$	$\mathcal{P}_t^{\text{att}}$	VP $Q\downarrow$	VP DRE $\uparrow$	VP VOS Avg.$\downarrow$	OP $Q\downarrow$	OP DRE $\uparrow$	OP VOS Avg.$\downarrow$
Distractor evidence									
$\mathcal{P}_t^{\text{sem}}$ only	✓			0.651	7.42	69.47	0.312	7.34	39.52
w/o $\mathcal{P}_t^{\text{sem}}$		✓	✓	0.553	6.55	63.38	0.215	4.56	31.74
w/o $\mathcal{P}_t^{\text{alt}}$	✓		✓	0.591	8.91	65.96	0.290	7.94	37.21
w/o $\mathcal{P}_t^{\text{att}}$	✓	✓		0.501	9.34	58.79	0.255	8.12	34.03
History and routing									
FIFO $\mathcal{E}$	✓	✓	✓	0.611	4.55	67.14	0.211	5.44	31.02
Uniform routing	✓	✓	✓	0.591	6.55	64.81	0.155	7.43	26.41
First-frame banks	✓	✓	✓	0.742	1.44	69.91	0.131	8.71	22.00
Current-frame banks	✓	✓	✓	0.581	7.40	58.34	–	–	–
Full CDS	✓	✓	✓	0.420	10.83	50.97	0.076	9.28	19.56

6. Ablation Studies

Unless varied, both settings use 10 optimization passes, with $\epsilon = 8/255$ for VP and $\epsilon = 4/255$ for OP. The default OP rollout uses $N = 20$ frames.

Across the attack objectives, mask suppression weakens both tasks but induces little drift. Displacement ($\mathcal{L}_{\text{disp}}$) redirects this failure toward spatial misdirection, while clutter corruption ($\mathcal{L}_{\text{clut}}$) strengthens the attack across both tasks. Their combination reduces Q to 0.420 under VP and 0.076 under OP. Removing mask suppression raises OP DRE but is substantially weaker in Q and VOS Avg. **For iteration length,** ten passes substantially outperform five in both settings. Fifteen further improves VP but gains

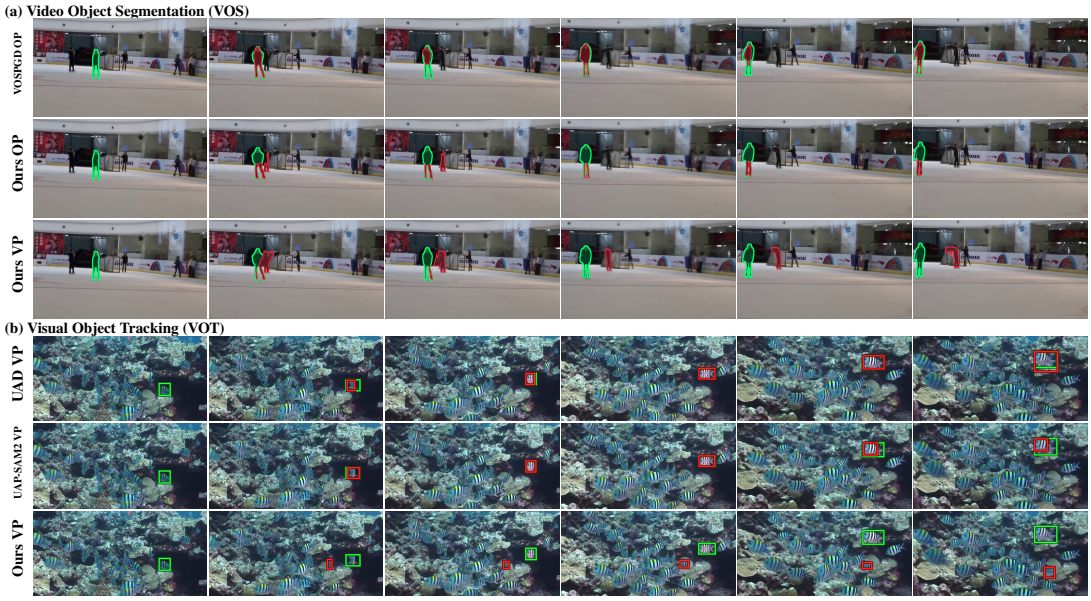


Figure 6. **Qualitative comparison of adversarial methods on (a) VOS and (b) VOT tasks.** Columns represent propagation from left to right, where the first column is the prompted frame. Green denotes the clean SAM2.1-L prediction, while red denotes its prediction under the corresponding perturbation and setting. Across both tasks, DistractAttack causes more severe and persistent failures than baselines.

only 0.005 in OP Q while worsening DRE and VOS performance. We therefore retain ten passes. **With increasing perturbation magnitude,** a larger budget strengthens the attack but permits more visible perturbations. Both Q and VOS Avg. decrease with ϵ, while VP DRE peaks at the selected $8/255$. **Across sources of distractor evidence,** semantic proposals provide broad coverage, while alternative masks and adversarial attention add decoder-specific cues. Removing any source weakens both tasks, and every reduced variant remains behind full CDS. **For historical banks and the update policy,** restricting $\mathcal{E}$ to the first or current frame weakens the attack. The current-frame variant applies only to VP because OP optimizes the initialization frame alone. FIFO replaces entries by arrival order, whereas uniform updates ignore consistency with existing distractors, so neither effectively preserves or prioritizes recurring evidence. Full CDS performs best by maintaining a persistent $\mathcal{E}$ and prioritizing proposals supported by its most effective entries. Supporting ablations are given in Appendix 9.4.

Defense Case Study. We evaluate complementary feature- and memory-level defenses against DistractAttack. Robust-SAM [27] adversarially adapts the singular values of selected convolutional weights. We further repurpose SAM2Long's attention modulation [9] to attenuate low-confidence memory keys and limit uncertain time steps and initialization errors. Robust-SAM and our memory defense raise Q from 0.420 to 0.515 and 0.476, respectively. Combined, they reach 0.532 Q and reduce DRE from 10.83% to 8.45%, but remain well below the clean Q of 0.836. Existing feature and memory controls therefore offer only partial protection. Details are provided in Appendix 10.

7. Conclusion

We examined how adversarial perturbations compromise target identity in SAM2. Although recurrent memory promotes temporal consistency, it can propagate perturbation-induced local ambiguity into persistent tracking failure. We introduced DistractAttack, a distractor-centric attack that exploits plausible competing regions rather than relying solely on foreground suppression. Its Counterfactual Distractor Search jointly follows clean and attacked trajectories, identifies model-supported distractors, and accumulates their evidence over time to sustain failure. Across VOT and VOS, DistractAttack remains effective under both perturbation settings and long temporal horizons, while generalizing and transferring across model scales and tracker variants. These results identify distractor-induced failure and initialization sensitivity as key vulnerabilities of SAM2-based trackers, motivating stronger robustness mechanisms in video foundation models.

Broader Societal Impact

This work provides new insight into the vulnerabilities of SAM2. We hope these findings can inform future efforts to better understand and mitigate such failure modes, contributing to safer and more reliable deployment.

References

[1] Naveed Akhtar, Ajmal Mian, Navid Kardan, and Mubarak Shah. Advances in adversarial attacks and defenses in computer vision: A survey. *IEEE Access*, 9:155161–155196, 2021. 1

[2] Basit Alawode, Yuhang Guo, Mehnaz Ummar, Naoufel Werghi, Jorge Dias, Ajmal Mian, and Sajid Javed. Utb180: A high-quality benchmark for underwater tracking. In *ACCV*, 2022. 6, 1

[3] Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris Coll-Vinent, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, et al. Sam 3: Segment anything with concepts. In *International Conference on Learning Representations*, pages 138846–138923, 2026. 2

[4] Ruixiang Chen, Guolei Sun, Yawei Li, Jie Qin, and Luca Benini. Him2sam: Enhancing sam2 with hierarchical motion estimation and memory optimization towards long-term tracking. In *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*, pages 276–291. Springer, 2025. 1, 2, 7

[5] Wei Chen and Yunqi Zhang. Csa: Crafting adversarial examples via content and style attacks. *Journal of Information Security and Applications*, 89:103974, 2025. 1, 2

[6] Xuesong Chen, Xiyu Yan, Feng Zheng, Yong Jiang, Shu-Tao Xia, Yong Zhao, and Rongrong Ji. One-shot adversarial attacks on visual tracking with dual attention. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10173–10182. IEEE, 2020. 1, 3, 6

[7] Zhigang Dai, Bolun Cai, and Junying Chen. Unimoco: unsupervised, semi-supervised and fully-supervised visual representation learning. In *2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 3099–3106. IEEE, 2022. 5

[8] Henghui Ding, Kaining Ying, Chang Liu, Shuting He, Xudong Jiang, Yu-Gang Jiang, Philip H. S. Torr, and Song Bai. Mosev2: A more challenging dataset for video object segmentation in complex scenes, 2025. 6, 1

[9] Shuangrui Ding, Rui Qian, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Yuhang Cao, Yuwei Guo, Dahua Lin, and Jiaqi Wang. Sam2long: Enhancing sam 2 for long video segmentation with a training-free memory tree. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 13614–13624. IEEE, 2025. 1, 2, 7, 8, 3

[10] Heng Fan, Hexin Bai, Liting Lin, Fan Yang, Peng Chu, Ge Deng, Sijia Yu, Harshit, Mingzhen Huang, Juehuan Liu, Yong Xu, Chunyuan Liao, Lin Yuan, and Haibin Ling. Lasot: A high-quality large-scale single object tracking benchmark, 2020. 6, 1

[11] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*, 2014. 6

[12] Jindong Gu, Hengshuang Zhao, Volker Tresp, and Philip HS Torr. Segpgd: An effective and efficient adversarial attack for evaluating and boosting segmentation robustness. In *European Conference on Computer Vision*, pages 308–325. Springer, 2022. 6, 2

[13] Qing Guo, Xiaofei Xie, Felix Juefei-Xu, Lei Ma, Zhongguo Li, Wanli Xue, Wei Feng, and Yang Liu. Spark: Spatial-aware online incremental attack against visual tracking, 2020. 1, 2

[14] Qing Guo, Ziyi Cheng, Felix Juefei-Xu, Lei Ma, Xiaofei Xie, Yang Liu, and Jianjun Zhao. Learning to adversarially blur visual object tracking, 2021. 1, 2

[15] Dongshen Han, Chaoning Zhang, Sheng Zheng, Chang Lu, Yang Yang, and Heng Tao Shen. Sam meets uap: Attacking segment anything model with universal adversarial perturbation, 2024. 1, 2

[16] Lianghua Huang, Xin Zhao, and Kaiqi Huang. Got-10k: A large high-diversity benchmark for generic object tracking in the wild. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(5):1562–1577, 2021. 6, 1

[17] Yihao Huang, Yue Cao, Tianlin Li, Felix Juefei-Xu, Di Lin, Ivor W Tsang, Yang Liu, and Qing Guo. On the robustness of segment anything. *arXiv preprint arXiv:2305.16220*, 2023. 1, 2

[18] Shuai Jia, Yibing Song, Chao Ma, and Xiaokang Yang. Iou attack: Towards temporally coherent black-box adversarial attack for visual object tracking. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6705–6714. IEEE, 2021. 1, 2

[19] Kaixun Jiang, Lingyi Hong, Zhaoyu Chen, Pinxue Guo, Zeng Tao, Yan Wang, and Wenqiang Zhang. Exploring the adversarial robustness of video object segmentation via one-shot adversarial attacks. In *Proceedings of the 31st ACM International Conference on Multimedia (MM'23)*, pages 8598–8607. ACM, 2023. 1, 2, 3, 6, 7

[20] Yan Jiang, Guisheng Yin, Ye Yuan, Jingjing Chen, and Zhipeng Wei. Cross-point adversarial attack based on feature neighborhood disruption against segment anything model. In *2024 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6. IEEE, 2024. 1, 2

[21] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020. 5

[22] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything, 2023. 1, 2

[23] Matej Kristan, Jiří Matas, Martin Danelljan, Michael Felsberg, Hyung Jin Chang, Luka Čehovin Zajc, Alan Lukežič, Ondrej Drbohlav, Zhongqun Zhang, Khanh-Tung Tran, et al. The first visual object tracking segmentation vots2023 challenge results. In *2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 1788–1810. IEEE, 2023. 2, 6, 1

[24] Ping Li, Yu Zhang, Li Yuan, Jian Zhao, Xianghua Xu, and Xiaoqin Zhang. Adversarial attacks on video object segmentation with hard region discovery. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(6):5049–5062, 2023. 1, 2, 6

[25] Siyuan Liang, Xingxing Wei, Siyuan Yao, and Xiaochun Cao. Efficient adversarial attacks for visual object tracking. In *European Conference on Computer Vision*, pages 34–50. Springer, 2020. 1, 2, 6, 7

[26] Yuchen Liu and Pengxu Wei. Cross-prompt adversarial attack on segment anything model. In *Proceedings of the 2024 12th international conference on communications and broadband networking*, pages 34–39, 2024. 1, 2

[27] Jiahuan Long, Zhengqin Xu, Tingsong Jiang, Wen Yao, Shuai Jia, Chao Ma, and Xiaoqian Chen. Robust sam: On the adversarial robustness of vision foundation models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 5775–5783, 2025. 1, 2, 6, 8, 3

[28] Jiahao Lu, Xingyi Yang, and Xinchao Wang. Unsegment anything by simulating deformation. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24294–24304. IEEE, 2024. 1, 2, 6, 7

[29] Kira Maag and Asja Fischer. Uncertainty-weighted loss functions for improved adversarial attacks on semantic segmentation. In *2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, page 3894–3902. IEEE, 2024. 5

[30] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. *arXiv preprint arXiv:1706.06083*, 2017. 6

[31] Jordi Pont-Tuset, Federico Perazzi, Sergi Caelles, Pablo Arbeláez, Alex Sorkine-Hornung, and Luc Van Gool. The 2017 davis challenge on video object segmentation. *arXiv preprint arXiv:1704.00675*, 2017. 6, 1

[32] Yi Qin, Rui Wang, Tao Huang, Tong Xiao, and Liping Jing. Sam encoder breach by adversarial simplicial complex triggers downstream model failures. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10624–10634. IEEE, 2025. 1, 2

[33] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. In *International Conference on Learning Representations*, pages 28085–28128, 2025. 1, 2, 6, 3

[34] Hao Ren, Shoudong Han, Huilin Ding, Ziwen Zhang, Hongwei Wang, and Faquan Wang. Focus on details: Online multi-object tracking with diverse fine-grained representation, 2023. 5

[35] Yifan Shen, Zhengyuan Li, and Gang Wang. Practical region-level attack against segment anything models. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 194–203. IEEE, 2024. 1, 2

[36] Ye Sun, Xin Wang, Jiaming Zhang, Yifeng Gao, Yixu Wang, Yifan Ding, Qixian Zhang, Henghui Ding, Xingjun Ma, and Yu-Gang Jiang. Darkllm: Learning language-driven adversarial attacks with large language models. *arXiv preprint arXiv:2605.18868*, 2026. 2

[37] Jovana Videnovic, Alan Lukezic, and Matej Kristan. A distractor-aware memory for visual object tracking with sam2. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24255–24264. IEEE, 2025. 1, 2, 6, 7

[38] Song Xia, Wenhan Yang, Yi Yu, Xun Lin, Henghui Ding, Ling-Yu Duan, and Xudong Jiang. Transferable adversarial attacks on sam and its downstream models, 2025. 1, 2

[39] Haodong Xiao, Wenbo Yu, Hao Fang, Shuoyang Sun, Bin Chen, Xuan Wang, and Shu-Tao Xia. Diffusion-based natural adversarial perturbations towards segment anything model. In *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 13637–13641. IEEE, 2026. 1, 2

[40] Ning Xu, Linjie Yang, Yuchen Fan, Dingcheng Yue, Yuchen Liang, Jianchao Yang, and Thomas Huang. Youtube-vos: A large-scale video object segmentation benchmark. *arXiv preprint arXiv:1809.03327*, 2018. 6, 1

[41] Qianxiong Xu, Lanyun Zhu, Chenxi Liu, Guosheng Lin, Cheng Long, Ziyue Li, and Rui Zhao. Samite: Position prompted sam2 with calibrated memory for visual object tracking. *arXiv preprint arXiv:2507.21732*, 2025. 1, 2, 4, 7

[42] Cheng-Yen Yang, Hsiang-Wei Huang, Wenhao Chai, Zhongyu Jiang, and Jenq-Neng Hwang. Samurai: Adapting segment anything model for zero-shot visual tracking with motion-aware memory. *arXiv preprint arXiv:2411.11922*, 2024. 1, 2, 7

[43] Chenshuang Zhang, Chaoning Zhang, Taegoo Kang, Donghun Kim, Sung-Ho Bae, and In So Kweon. Attack-sam: Towards attacking segment anything model with adversarial examples, 2023. 1, 2, 5, 6

[44] Zongmin Zhang, Zhen Sun, Yifan Liao, Wenhan Dong, Xinlei He, Xingshuo Han, Shengmin Xu, and Xinyi Huang. Backdoor attacks on prompt-driven video segmentation foundation models. *arXiv preprint arXiv:2512.22046*, 2025. 2

[45] Sheng Zheng, Chaoning Zhang, and Xinhong Hao. Black-box targeted adversarial attack on segment anything (sam). *IEEE Transactions on Multimedia*, 27:1901–1913, 2024. 1, 2

[46] Ziqi Zhou, Yufei Song, Minghui Li, Shengshan Hu, Xianlong Wang, Leo Yu Zhang, Dezhong Yao, and Hai Jin. Darksam: Fooling segment anything model to segment nothing. *Advances in Neural Information Processing Systems*, 37:49859–49880, 2024. 1, 2

[47] Ziqi Zhou, Yifan Hu, Yufei Song, Zijing Li, Shengshan Hu, Leo Yu Zhang, Dezhong Yao, Long Zheng, and Hai Jin. Vanish into thin air: Cross-prompt universal adversarial attacks for sam2. *Advances in Neural Information Processing Systems*, 38:142839–142861, 2026. 2, 5, 6

Distractors are the Real Enemy: Visual Distraction Adversarial Perturbations on SAM2-Based Video Tracking and Segmentation

Supplementary Material

- **8. Implementation Details**
  - 8.1. Attack Optimization
  - 8.2. CDS Configuration
  - 8.3. Datasets and Metrics
  - 8.4. Baseline Adaptations
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  - 9.2. VOS Transferability
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- **10. Defense Study**
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  - 10.2. Training-Free Attention Modulation

8. Implementation Details

This section gives the settings used to reproduce DistractAttack. We follow the VP and OP definitions in Sec. 4. The victim model and its tracker state remain differentiable, while all model parameters are frozen.

8.1. Attack Optimization

Shared setup. We use the official pretrained SAM 2.1-L checkpoint [33]. Frames follow the official SAM 2 preprocessing and are resized to 1024×1024. The perturbation is initialized from $\delta \sim \mathcal{U}(-\epsilon, \epsilon)$, with ϵ defaulting to 8/255 and 4/255, for VP and OP respectively, and updated with the projected signed gradient rule in Eq. (18). After every update, δ is projected onto the ℓ_∞ ball of radius ϵ. When composing an attacked frame, $x + \delta$ is clipped to $[0, 1]$ before normalization.

Video perturbation. VP optimizes one perturbation for each video. The same perturbation is applied to every frame in its fixed window. Clean and attacked trajectories are propagated together during each pass. Their detached banks provide the temporal context used by the attack losses.

One-shot perturbation. OP changes only the prompted frame. Later frames remain clean and provide evidence for the target and effective-distractor banks. The clutter objective acts on the perturbed frame 0 encoder features, while later clean frames contribute detached evidence to the banks. Neither setting uses ground truth beyond the initialization prompt during optimization.

8.2. CDS Configuration

Banks. The target bank stores at most 20 foreground and background prototype pairs. Later entries require predicted-IoU and objectness scores above 0.5 and 0, respectively. We use a novelty threshold of $\tau_{\text{nov}} = 0.98$. These checks prevent closely repeated observations from filling the bank. The effective-distractor bank has capacity $K = 10$. Its merge threshold is $\theta_{\text{merge}} = 0.85$ and its EMA rate is $\eta = 0.1$. We set the routing temperature to $\tau_w = 0.10$ and the clutter-loss contrastive temperature to $\tau_c = 0.07$.

Loss. A grid search over $\{0.25, 0.5, 1.0\}$ for each loss weight selected $\lambda_M = \lambda_D = \lambda_C = 1.0$. The global and local clutter weights are $\eta_G = 1.0$ and $\eta_L = 1.0$.

8.3. Datasets and Metrics

Visual object tracking. The VOT set contains 104 randomly sampled sequences from LaSOT [10], UTB180 [2], GOT-10k [16], and LaSOT-ext [10]. The LaSOT and UTB180 subset follows the distractor-distilled DiDi selection [37]. GOT-10k and LaSOT-ext add broader tracking conditions including videos with no distractors. Each source dataset contributes 22-31 randomly sampled sequences. Each sequence contributes a long 100-frame window, in contrast to 15 frame window used by published UAP-SAM2. None of these evaluation splits was used to train the original SAM 2 model.

Video object segmentation. The VOS set contains 100 videos. We sample 25 videos from each source and select one target object. DAVIS 2017 [31] provides accurate boundaries. YouTube-VOS 2018 validation [40] adds broad appearance and category variation. The MOSEv2 [8] contains crowded scenes and severe occlusion. SA-V test [33] covers motion and disappearance. Similarily, each video is 100 frame long. None of these evaluation splits was used to train the original SAM 2 model.

VOT metrics. Each predicted mask is converted to its tight box, with an empty mask denoting target absence. Let $\mathcal{I}_s$ contain the evaluated frames of sequence s, excluding initialization and unavailable annotations, and let $N_s = |\mathcal{I}_s|$. We denote the predicted box, ground-truth box, and their IoU by $\widehat{B}_{s,t}$, $B_{s,t}$, and $o_{s,t}$, respectively. We follow VOTS2023 [23] for all our VOT metrics. We use $o_{s,t} = 1$ when both boxes are empty and $o_{s,t} = 0$ when only one is empty. Tracking quality is

$$Q = \frac{1}{S} \sum_{s=1}^{S} \frac{1}{N_s} \sum_{t \in \mathcal{I}_s} o_{s,t}. \tag{20}$$

Let $\mathcal{V}_s \subseteq \mathcal{I}_s$ contain the visible-target frames. Within this set, T_s counts predictions with nonzero overlap, D_s

counts non-empty predictions with zero overlap, and M_s counts empty predictions. The remaining metrics are

$$\mathrm{Acc} = \frac{1}{S}\sum_{s=1}^{S}\frac{1}{T_s}\sum_{\substack{t\in\mathcal{V}_s\\ o_{s,t}>0}} o_{s,t}, \tag{21}$$

$$\mathrm{Rob} = \frac{1}{S}\sum_{s=1}^{S}\frac{T_s}{T_s+D_s+M_s}, \tag{22}$$

$$\mathrm{DRE} = \frac{1}{S}\sum_{s=1}^{S}\frac{D_s}{T_s+D_s+M_s}, \tag{23}$$

$$\mathrm{NRE} = \frac{1}{S}\sum_{s=1}^{S}\frac{M_s}{T_s+D_s+M_s}. \tag{24}$$

Tables report DRE and NRE as percentages. A stronger attack lowers Q, accuracy, and robustness while increasing DRE and NRE.

VOS metrics. Following SAM 2 [33], we report the standard $\mathcal{J}\&\mathcal{F}$ score, which averages mask IoU and contour accuracy over annotated non-initialization frames and then over videos. Lower is stronger.

8.4. Baseline Adaptations

Common protocol. We retain each baseline's defining objective and apply it under the same VP or OP constraint as DistractAttack. VP aggregates each method's native loss over the sequence. OP perturbs only the initialization frame. When a method requires a reference after frame 0, we use detached clean predictions rather than future annotations. All methods share the same preprocessing, prompts, temporal windows, random seeds, ℓ_∞ bounds.

Image-based attacks. We adapt AttackSAM, DarkSAM, CPA, SegPGD, and UAD from static image segmentation to the recurrent setting. AttackSAM [43] applies its ClipMSE objective to the selected SAM 2 mask prediction. DarkSAM [46] retains its semantic-decoupling and wavelet-texture terms. CPA [27] retains its negative-feature objective over the top responses shared by point and box prompts. SegPGD [12] treats each SAM 2 output as binary segmentation, uses the detached clean mask as its pseudo-label, and retains its iteration-dependent weighting schedule. UAD [28] preserves its two-stage feature objective with deformation fields. VP aggregates this objective across the rollout, while OP applies it to the initialization entering memory.

UAP-SAM2. UAP-SAM2 [47] combines target scanning, semantic confusion, encoder feature shift, and temporal feature misalignment. We retain these terms while changing setting to VP and OP.

One-shot VOS attacks. ARA-Attack [24] and VOSPGD [19] remain in their native one-shot setting. ARA-Attack recomputes the initialization gradient map and applies its hard-region learner. VOSPGD writes the perturbed prompt frame to memory and replays its clean counterpart as a memory-conditioned query. For VOT, the clean mask obtained from the initialization box provides the binary reference, while VOS uses the given initialization mask.

Table 3. Per-dataset VOT results extending Table 1

		VP			OP		
Dataset	Seq.	$Q\downarrow$	DRE (%)↑	NRE (%)↑	$Q\downarrow$	DRE (%)↑	NRE (%)↑
Distractor-heavy							
LaSOT	22	0.443	11.02	14.44	0.010	8.47	71.95
UTB180	22	0.385	18.37	10.01	0.156	12.18	47.84
Generic							
GOT-10k	31	0.470	4.52	14.96	0.049	9.76	70.93
LaSOT-ext	29	0.377	11.70	22.01	0.095	7.18	69.04
Total	**104**	**0.420**	**10.83**	**15.77**	**0.076**	**9.28**	**65.73**

9. Supplementary Evaluation

9.1. Per-Dataset VOT: Generic vs. Distractor-heavy

To complement the aggregate VOT evaluation in Tab. 1, we report results separately on LaSOT, UTB180, GOT-10k, and LaSOT-ext (See Tab. 3). All datasets use the same fixed windows, initialization protocol, and attack settings as the main evaluation. Results are macro-averaged within each dataset with equal weight per sequence, while the combined row covers all sequences. This breakdown examines consistency across datasets with some variation shown. GOT-10k and LaSOT-ext cover broader generic tracking conditions and are not restricted to scenes with explicit competing objects, yet exhibit comparable degradation. Thus, the attack does not require a literal distractor in every frame: CDS exploits model-supported off-target regions when available, while foreground suppression remains effective otherwise.

9.2. VOS Transferability

We evaluate VP transfer across SAM2.1, DAM4SAM, SAMURAI, SAMITE, and SAM2Long on DAVIS 2017. All variants use the SAM2.1-L checkpoint. The consistently low off-diagonal scores in Fig. 7 demonstrate strong transfer across tracker variants. DistractAttack remains effective despite their different tracking and memory designs, revealing a shared vulnerability across SAM 2-based trackers.

9.3. Additional Qualitative Results

Tracker variants. As shown in Fig. 8, we compare identical frames and prompts across DAM4SAM, SAMURAI, HiM2SAM, SAMITE, and SAM2Long on VOT VP. The examples complement the aggregate results in the main paper.

Additional method comparisons As shown in Fig. 9 and Fig. 10, we extend the qualitative comparison with FAN,

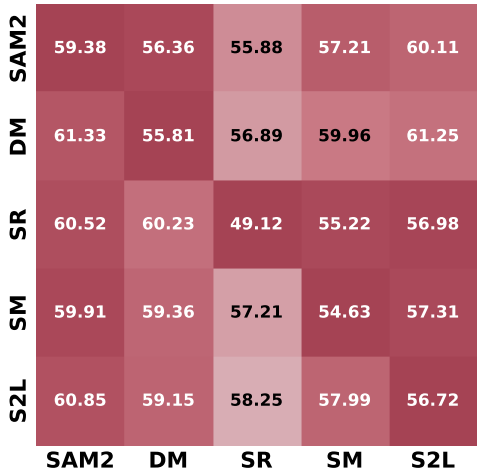

Figure 7. **Cross-variant VOS transferability under VP.** Each row denotes the source tracker used to optimize the perturbation, while each column denotes the evaluated target tracker. Lower $\mathcal{J}\&\mathcal{F}$ indicates a stronger attack.

Table 4. Effect of the signed-gradient step size. Arrows indicate a stronger attack.

Setting	Step	$Q\downarrow$	DRE%↑	VOS Avg.↓
VP	1/255	0.453	6.87	66.51
	2/255	**0.420**	**10.83**	**50.97**
	4/255	0.445	5.67	53.09
OP	1/255	**0.076**	9.28	**19.56**
	2/255	0.122	**10.23**	21.67

Table 5. Ablation of uniform and distractor-weighted $\mathcal{L}_{\text{disp}}$. Arrows indicate a stronger attack.

Setting	Displacement	$Q\downarrow$	DRE%↑	VOS Avg.↓
VP	Unweighted	0.450	2.51	54.11
	Bank-weighted	**0.420**	**10.83**	**50.97**
OP	Unweighted	0.145	1.43	24.13
	Bank-weighted	**0.076**	**9.28**	**19.56**

SegPGD, DarkSAM, UAD and UAP-SAM2 on VOT OP and VOS VP.

Model size variants. As shown in Fig. 11, we next compare the Tiny, Small, Base+, and Large sizes of SAM 2.1 on the same sequences.

Additional and more clear qualitative comparisons and temporal visualizations are provided in the accompanying supplementary video.

9.4. Additional Ablations

Step Size. As in Tab. 4, we fix the perturbation bound and optimization passes and vary only the signed-gradient step size. Under VP, $2/255$ gives the strongest overall degradation ($Q = 0.420$, DRE$= 10.83\%$, VOS Avg.$= 50.97$), while under OP, $1/255$ performs best overall ($Q = 0.076$, VOS Avg.$= 19.56$). These settings are therefore used as the default step sizes.

Unweighted versus weighted displacement. As in Tab. 5, we isolate the weighting in $\mathcal{L}_{\text{disp}}$ while retaining the same proposal set, effective-distractor bank, $\mathcal{L}_{\text{mask}}$, and $\mathcal{L}_{\text{clut}}$. The unweighted variant assigns $1/|\mathcal{P}_t|$ to every active proposal as in Eq. (12). The weighted variant uses the bank-supported w_t^j in Eqs. (13) and (14). Thus, the comparison changes only how foreground displacement is routed.

Table 6. Defense evaluation on SAM 2.1-L under VP and OP DistractAttack. The attack is recomputed against each defense.

Method	VP Q	VP DRE%	VP NRE%	OP Q	OP DRE%	OP NRE%
Clean reference	0.836	0.68	0.42	0.836	0.68	0.42
DistractAttack	0.420	10.83	15.77	0.076	9.28	65.73
Memory modulation [9]	0.476	9.52	10.33	0.101	10.55	60.21
Robust-SAM [27]	0.515	9.01	10.12	0.129	3.45	54.45
Both [9, 27]	0.532	8.45	10.42	0.133	4.51	55.45

10. Defense Study

The main paper tests a feature-level defense and a memory-level defense. We describe both adaptations here. The attack remains white box and is recomputed against each defended model.

10.1. Robust-SAM

Robust-SAM [27] adapts the singular values of selected convolutional weights. We apply this parameterization to the convolutional weights in the final image-encoder block of SAM 2.1-L. Each selected weight is decomposed as $W = UPV^\top$, where U and V remain frozen and only the diagonal matrix P is optimized. All other SAM 2 parameters remain fixed.

We train a separate defended model for each perturbation setting using CPA adversarial examples generated from 5,000 videos in the SA-V training split [33]. The training bound matches the corresponding evaluation setting, with $\epsilon = 8/255$ for VP and $\epsilon = 4/255$ for OP. Each model is adapted for 15 epochs using a batch size of 64 and Adam with a learning rate of 10^{-3}. Thus, the VP and OP results use independently adapted checkpoints rather than a single jointly trained defense.

10.2. Training-Free Attention Modulation

SAM2Long [9] improves memory cross-attention by weighting historical memories according to their object-confidence scores. We isolate this mechanism from the original memory-tree formulation and apply it directly to SAM 2.1-L, without additional training or learnable parameters. For the $N+1$ memory entries, we first construct linearly spaced weights between w_{low} and w_{high}:

$$w_i^{\text{std}} = w_{\text{low}} + \frac{i-1}{N}\,(w_{\text{high}} - w_{\text{low}})\,, \qquad i = 1, \ldots, N{+}1. \tag{25}$$

We then sort the associated confidence scores in ascending order, $o_{I_1} \le \cdots \le o_{I_{N+1}}$, and assign larger weights to memories with higher scores. The corresponding memory

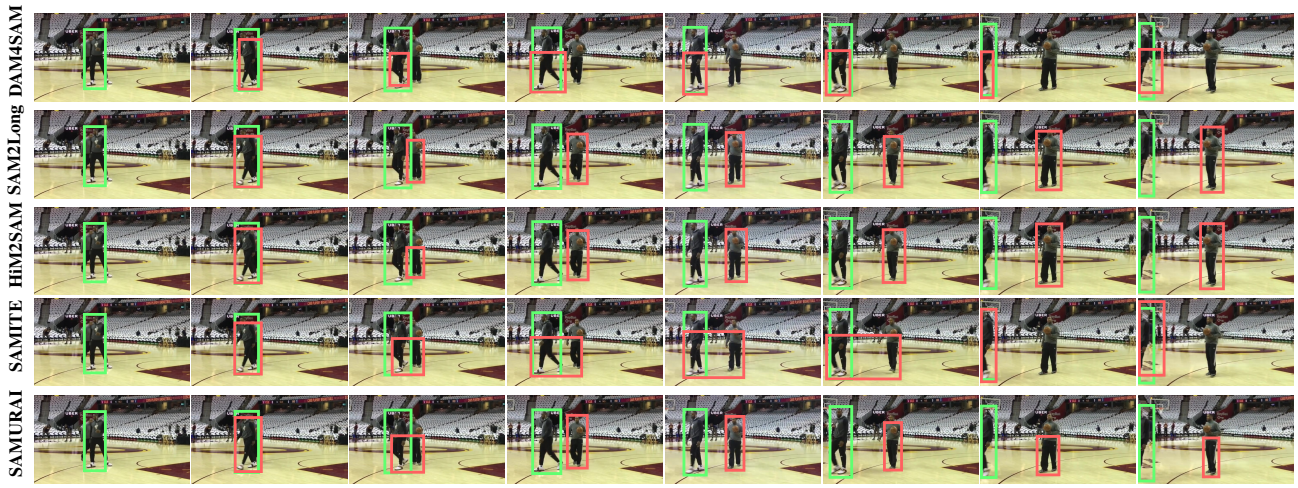

Figure 8. Qualitative comparison of perturbed tracker variants. Columns represent propagation from left to right, where the first column is the prompted frame. Green and red denote the clean reference and perturbed predictions, respectively.

keys are modulated before cross-attention as

$$w_{I_i} = w_i^{\mathrm{std}}, \qquad \widetilde{\mathbf{M}}_{I_i} = w_{I_i}\mathbf{M}_{I_i}, \qquad i = 1, \ldots, N+1. \tag{26}$$

Unlike SAM2Long, we adopt a wider weighting interval of $[0.75, 1.25]$ to increase the modulation strength, thereby amplifying the distinction between low- and high-confidence memories. Thus, reliable high-confidence memories contribute more strongly to the memory attention, while low-confidence entries are naturally suppressed.

As shown in Tab. 6, both defenses recover part of the lost tracking quality. Robust-SAM gives the larger individual gain. Their combination reaches the strongest defended result, yet it remains well below clean tracking. Existing feature and memory controls therefore reduce the failure without removing it. A stronger defense may need to estimate identity reliability before each memory is read.

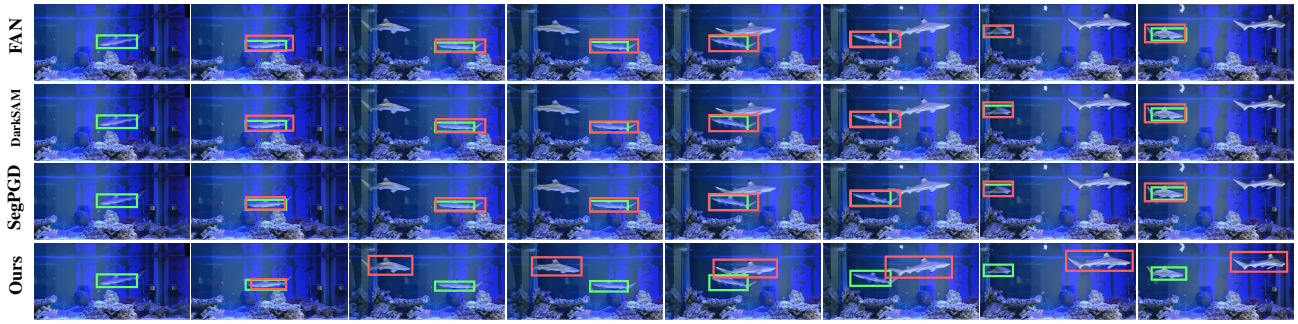

Figure 9. Qualitative comparison of VOT OP methods. Columns represent propagation from left to right, where the first column is the prompted frame. Green and red denote the clean reference and perturbed predictions.

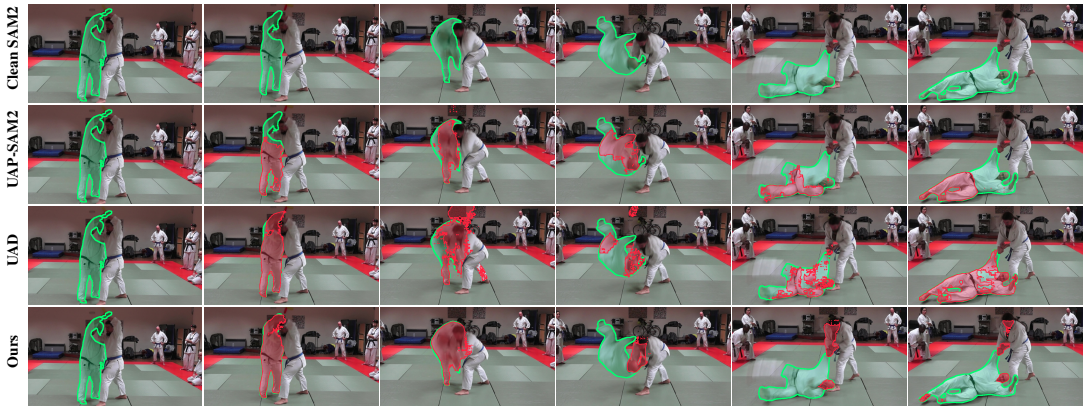

Figure 10. Qualitative comparison of VOS VP methods. Green denotes the Clean SAM2 reference prediction, while red denotes the perturbed prediction.

Figure 11. Qualitative comparison of perturbed SAM2.1 model-size variants on VOT VP. Columns represent propagation from left to right, where the first column is the prompted frame. Green and red denote the clean reference and perturbed predictions.